

```
sn@bharatha:~$ leave 0312
Alarm set for Tue Dec  8 15:12. (pid 15361)
sn@bharatha:~$ You have to leave in 5 minutes.
```

Figure 2: *leave* output

```

sn@ubuntu:~/Documents$ ls -la
total 120
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 .
drwxr-xr-x 16 sn sn 4096 Nov 15 11:25 ..
-rw-r--r--  1 sn sn  176 Nov 15 11:25 cpgrs
-rw-r--r--  1 sn sn  176 Nov 15 11:25 desktop
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 Documents
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 Downloads
-rw-r--r--  1 sn sn  176 Nov 15 11:25 duhalepr
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 Music
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 Pictures
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 Public
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 scripts
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 Templates
-rw-r--r--  1 sn sn  176 Nov 15 11:25 trial
drwxr-xr-x  2 sn sn 4096 Nov 15 11:25 Videos
-rw-r--r--  1 sn sn  176 Nov 15 11:25 comments.txt
-rw-r--r--  1 sn sn  176 Nov 15 11:25 examples.desktop
-rw-r--r--  1 sn sn  176 Nov 15 11:25 fl-
-rw-r--r--  1 sn sn  176 Nov 15 11:25 fl-
-rw-r--r--  1 sn sn  176 Nov 15 11:25 fl-

```

Figure 3: The *ranger* command output

```

sn@bharath:
File Edit View Search Terminal Help
sn@bharath:~/home/js/cprg/js/kunofdigits.c
# 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
#include <stdio.h>
int main()
{
    int n, i, sum, rem, digit;

    printf("\n enter the integer: ");
    scanf("%d", &n);

    sum = 0;
    rem = n % 10;
    while (n > 0)
    {
        digit = n % 10;
        printf("digit = %d\n", digit);
        sum = sum + digit;
        n = n / 10;
    }

    printf("\n sum of digits = %d\n", sum);
}

```

Figure 4: The *ranger* command with a view of the selected text file

## The *ranger* utility

The *ranger* utility is a text based file manager, written in Python, and it uses the *ncurses* library to provide an interface for the file system. The user interface is minimalistic and the navigation is very easy. It is available in most of the popular Linux distributions.

The utility can be installed in an Ubuntu system by the following command at the prompt:

```
$sudo apt-get install ranger
```

The file system can be navigated with this utility. It displays a three-column layout, with the left pane showing the current directory's parent directory, the middle column showing the current directory and the third pane showing a preview of the file. Many of the key bindings are the same as in the `vi` editor. The file system can be quickly navigated by using the keyboard, with the left (arrow) key used for going to the parent directory, the right (arrow) key to enter the selected

```
File Edit View Terminal Tabs Help
click.net:443 shiva FIRST_UP_PARENT/192.168.11.20 -
Dec 15 10:11:16 2015.387 479 192.168.1.106 TCP_MISS/000 491 CONNECT xns-ur1-go
.kaspersky-labs.com:443 rajivc FIRST_UP_PARENT/192.168.11.20 -
Dec 15 10:11:19 2015.347 318738 192.168.0.60 TCP_MISS/000 15903 CONNECT encrypted
.com:443 shiva FIRST_UP_PARENT/192.168.11.20 -
Dec 15 10:11:23 2015.346 18720 192.168.3.153 TCP_MISS/000 966 CONNECT www.google
.com:443 bhargap FIRST_UP_PARENT/192.168.11.20 -
Dec 15 10:11:24 2015.536 22085 192.168.3.153 TCP_MISS/000 776 CONNECT xns-crypto
-ur1-go.kaspersky-labs.com:443 bhargap FIRST_UP_PARENT/192.168.11.20 -
Dec 15 10:11:25 2015.536 22085 192.168.3.153 TCP_MISS/000 35146 CONNECT ap1s.go
gle.com:443 shiva FIRST_UP_PARENT/192.168.11.20 -
```

Figure 5: The *multitail* command output

[illegible]

Figure 6: The *multitail* command with column-wise display

directory, and up and down arrow keys used to browse the current directory.

The column at the right extreme displays a preview of the file. The default programs are used to display files. The *less* utility is used to display a text file, zipped files are uncompressed and displayed, the PDF files are displayed as text files and the image files are displayed using coloured ASCII art, though the clarity may not be good.

*Ranger* also includes a command interface, which helps you execute commands and manipulate files from within it. Several other options and features can be found by using the *man* command or from online sources.

## The *multitail* command

The *multitail* command is far better compared to the *tail* command for monitoring files. It allows the administrator to monitor multiple files and is particularly useful for monitoring the log files. One additional feature is that it allows you to monitor the output of command(s) as well. Written using *ncurses*, it creates multiple windows for monitoring the files and works in most of the popular Linux distributions.

The utility can be installed in an Ubuntu system by the following command at the prompt:

```
$sudo apt-get install multitail
```